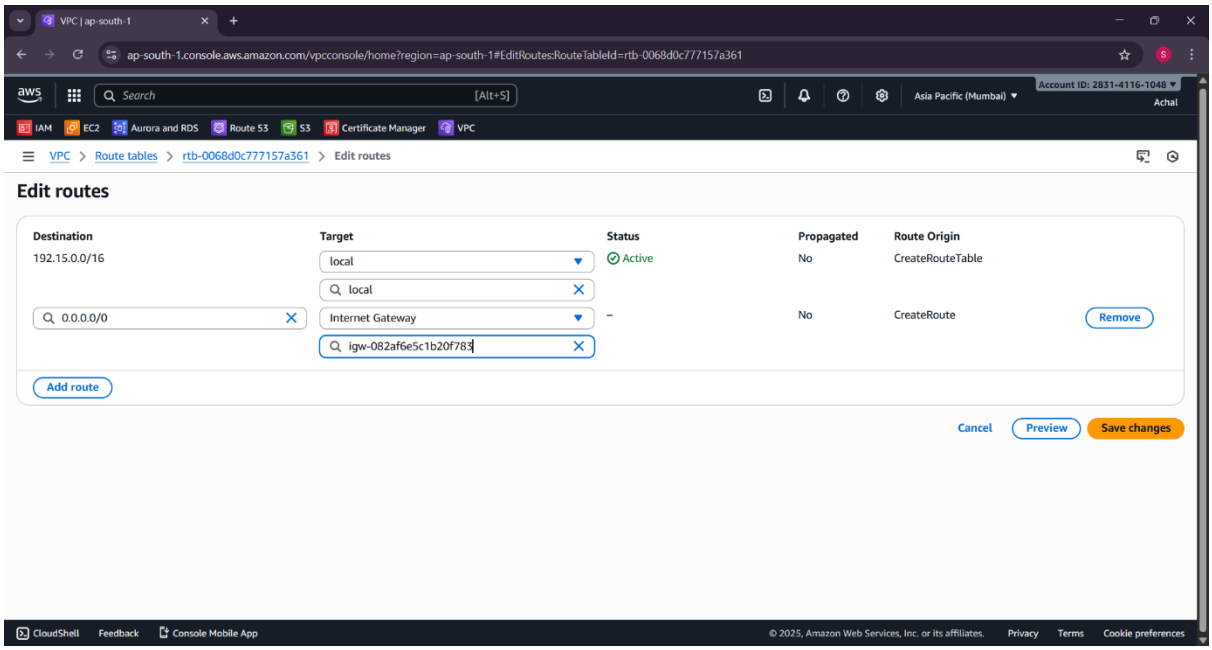
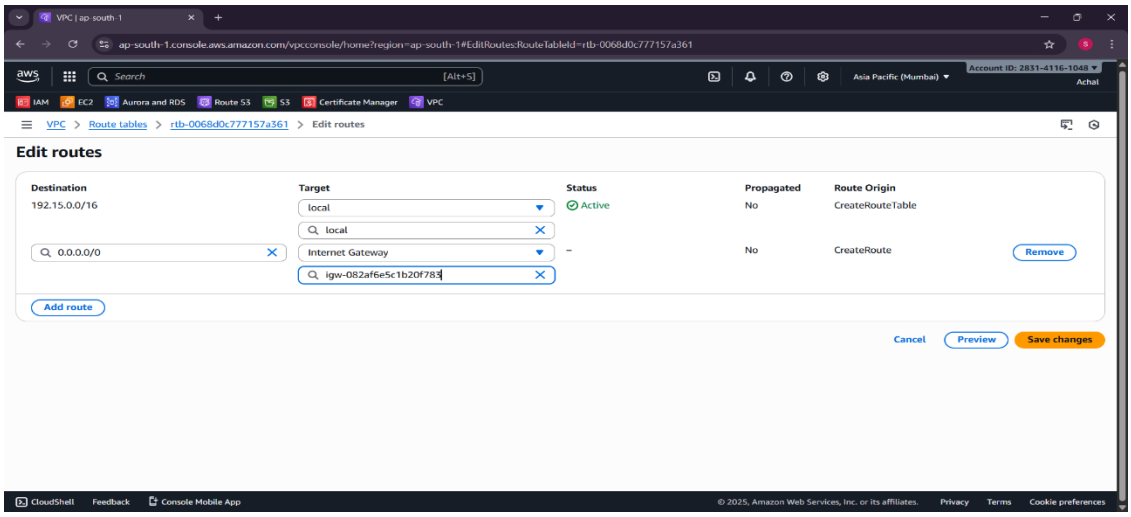
Task 20

- Host dynamic application and If I hit the Jump server IP, the application server should be accessible, and the database should be stored within the another vpc in different region.

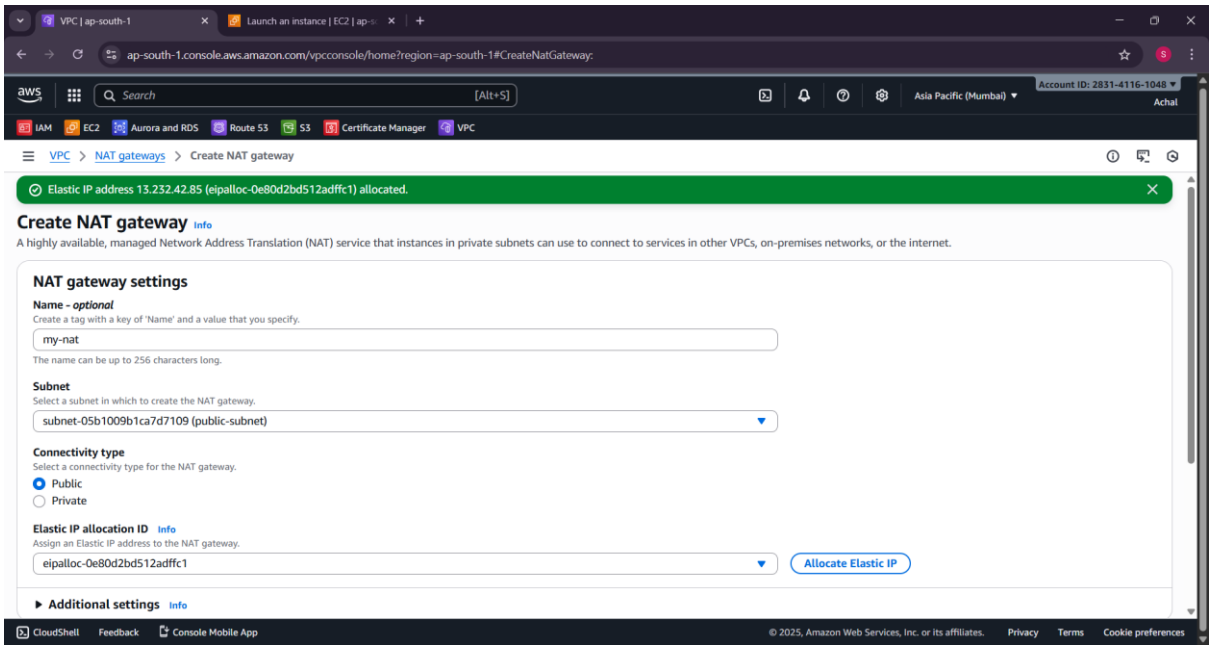
-Step 1: Create VPCa
-Step2: Create 2 Subnet One should be Public and one should be Private
-step3: Create Internet Gateway and attach to VPCa to provide internet and associate public subnet .

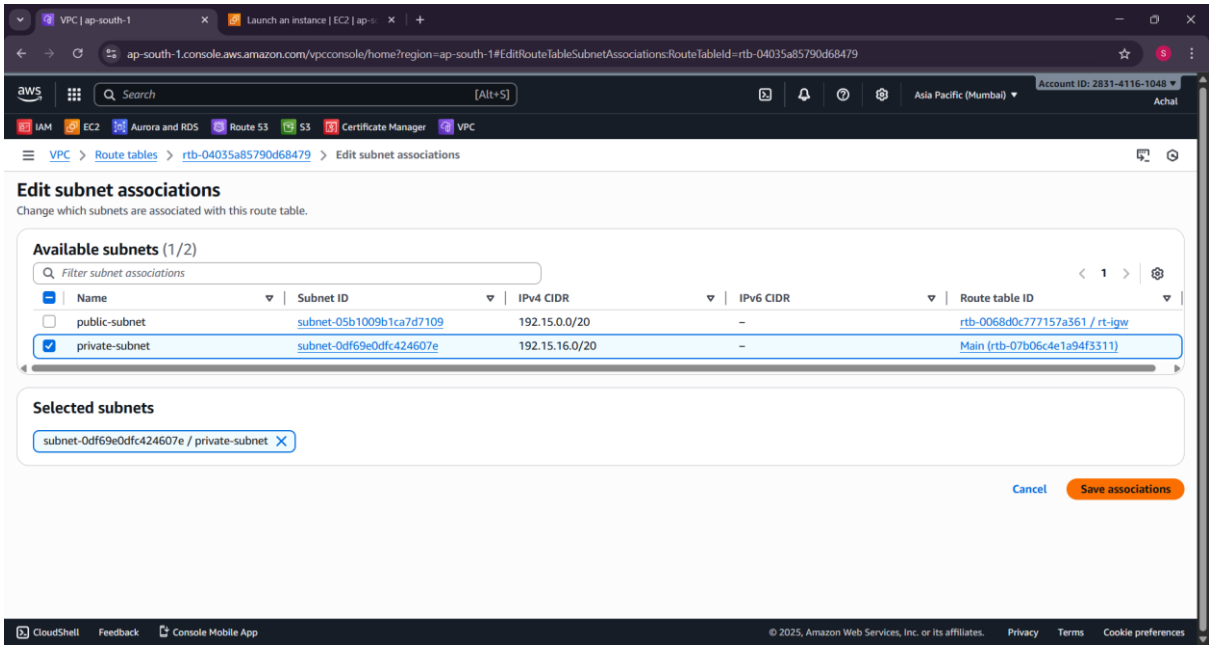


-step4: Create Route Table and route internet gateway in the route table and associate Public subnet into that route table.



-step 5: Create ec2 instance/jump server in public subnet with auto assigned public ip should be enable. Create nat gateway to provide internet to the application server. Create one route table and associate private subnet in the route table of the nat gateway .





-step 6: Create ec2 instance /application server in private subnet with auto assigned public ip should be disable.

-step 7: Copy key from local machine to jump server to take access of the application server.

CMD – scp -i key-name(authentication) key-name(file) ec2-user@publicip:/home/ec2-user

-step8: Connect jump server on git bash terminal

yum Install nginx

systemctl start nginx.service

systemctl enable nginx.service

-step 9: Install tomcat on the application server from the google

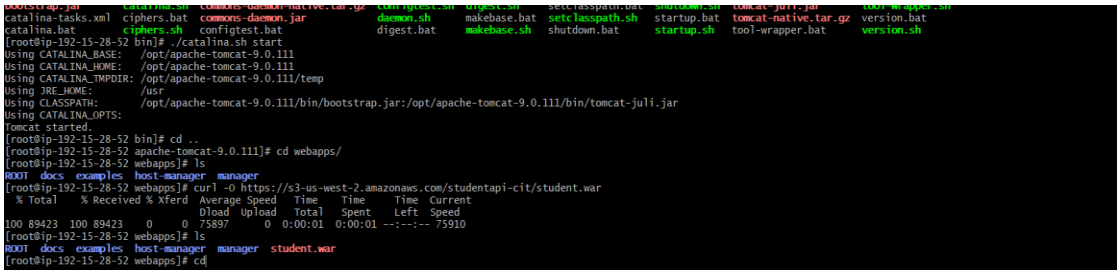
Curl -O <link of tomcat 9>

Then inside that tomcat folder bin folder will show enter inside bin folder and start Catalina.sh file then tomcat will start

tomcatfolder/bin/

CMD - ./Catalina.sh start

-step 10: In the tomcat folder inside the webapp folder stored our Application – student.war inside the webapp folder curl -O https://s3-us-west-2.amazonaws.com/studentapi-cit/student.war



-step 11: Do some changes in the nginix.conf file for whenever we hit Public ip of jump server then request will go towords the Application server.

In the jump server vim etc/nginx/nginx.conf

location / {

proxy_pass http://privateIPoftomcat:8080/student/;

}

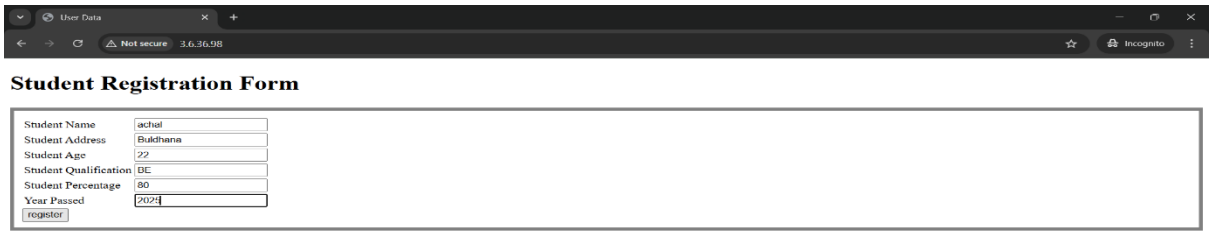
Do that changes in the configuration file of the nginx.

```
root@ip-192-15-9-68:~
# For more information on configuration, see:
#   * Official English Documentation: http://nginx.org/en/docs/
#   * Official Russian Documentation: http://nginx.org/ru/docs/

user nginx;
worker_processes auto;
error_log /var/log/nginx/error.log notice;
pid /run/nginx.pid;

# Load dynamic modules. See /usr/share/doc/nginx/README.dynamic.
include /usr/share/nginx/modules/*.conf;

events {
    worker_connections 1024;
}

http {
    log_format  main  '$remote_addr - $remote_user [$time_local] "$request" '
                      '$status $body_bytes_sent "$http_referer" '
                      '"$http_user_agent" "$http_x_forwarded_for"';

    access_log  /var/log/nginx/access.log  main;

    sendfile            on;
    tcp_nopush          on;
    keepalive_timeout   65;
    types_hash_max_size 4096;

    include             /etc/nginx/mime.types;
    default_type        application/octet-stream;

    # Load modular configuration files from the /etc/nginx/conf.d directory.
    # See http://nginx.org/en/docs/ngx_core_module.html#include
    # for more information.
    include /etc/nginx/conf.d/*.conf;

    server {
        listen       80;
        listen       [::]:80;
        server_name  _;
        root         /usr/share/nginx/html;

        # Load configuration files for the default server block.
        include /etc/nginx/default.d/*.conf;

        error_page 404 /404.html;
        location = /404.html {
        }
        location / {
        proxy_pass http://192.15.28.52:8080/student/;
}
        error_page 500 502 503 504 /50x.html;
        location = /50x.html {
        }
    }

# Settings for a TLS enabled server.
                                                                    50,38         Top
```

After this all changes restart nginx.

-step 12: After this all changes when we hit public ip of the jump server then

The request is forword towards the application server.



Sorry! unable to save record

-step 13: Create VPCb and create one subnet

-step 15: create ec2instance in VPC which is a db instance

-step 14: install mariadb105* then start and enable

-step 15: Connet that ec2-instance on the git bash then configure database

mysql -h <endpoint> -u <username> -p<password>

$ create database studentapp;

$ use studentapp;

$ create table by using below query

CREATE TABLE if not exists students(student_id INT NOT NULL AUTO_INCREMENT,

student_name VARCHAR(100) NOT NULL,

student_addr VARCHAR(100) NOT NULL,

student_age VARCHAR(3) NOT NULL,

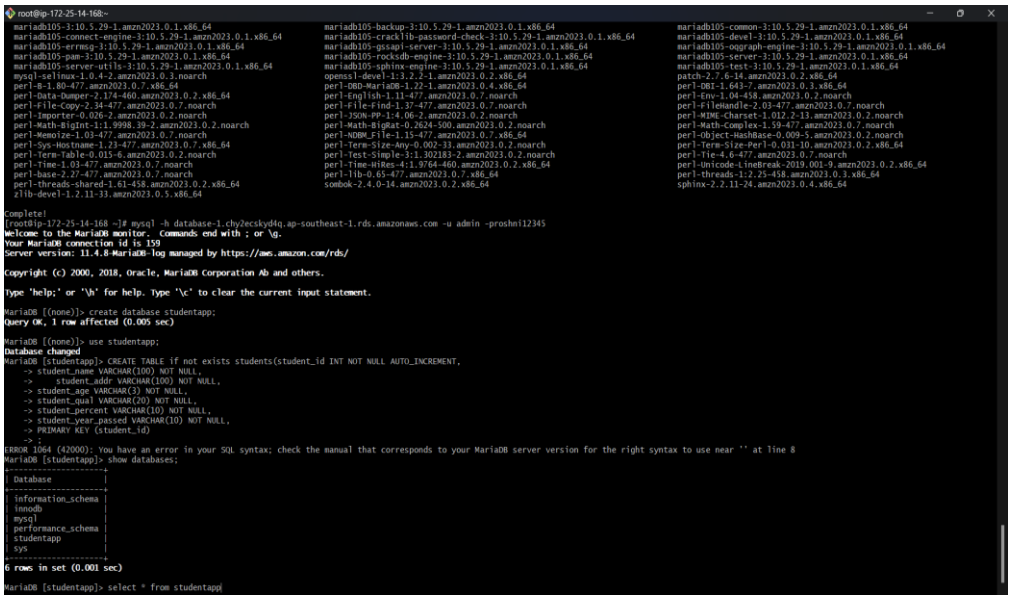
student_qual VARCHAR(20) NOT NULL,

student_percent VARCHAR(10) NOT NULL,

student_year_passed VARCHAR(10) NOT NULL,

PRIMARY KEY (student_id)

);



-step 16: After that in the application server inside the tomcat folder inside lib

Add connector file

curl -O https://s3-us-west-2.amazonaws.com/studentapi-cit/mysql-connector.jar

```
root@ip-192-15-28-52:/opt/apache-tomcat-9.0.111/lib
apache-tomcat-9.0.111/webapps/manager/xform.xsl
apache-tomcat-9.0.111/bin/catalina.sh
apache-tomcat-9.0.111/bin/ciphers.sh
apache-tomcat-9.0.111/bin/configtest.sh
apache-tomcat-9.0.111/bin/daemon.sh
apache-tomcat-9.0.111/bin/digest.sh
apache-tomcat-9.0.111/bin/makebase.sh
apache-tomcat-9.0.111/bin/setclasspath.sh
apache-tomcat-9.0.111/bin/shutdown.sh
apache-tomcat-9.0.111/bin/startup.sh
apache-tomcat-9.0.111/bin/tool-wrapper.sh
apache-tomcat-9.0.111/bin/version.sh
[root@ip-192-15-28-52 ~]# cd /opt
[root@ip-192-15-28-52 opt]# ls
apache-tomcat-9.0.111  aws
[root@ip-192-15-28-52 opt]# cd apache-tomcat-9.0.111/
[root@ip-192-15-28-52 apache-tomcat-9.0.111]# ls
BUILDING.txt  CONTRIBUTING.md  LICENSE  NOTICE  README.md  RELEASE-NOTES  RUNNING.txt  bin  conf  lib  logs  temp  webapps  work
[root@ip-192-15-28-52 apache-tomcat-9.0.111]# cd bin
[root@ip-192-15-28-52 bin]# ls
bootstrap.jar       catalina.sh  commons-daemon-native.tar.gz  configtest.sh  digest.sh      setclasspath.bat  shutdown.sh  tomcat-juli.jar       tool-wrapper.sh
catalina-tasks.xml  ciphers.bat  commons-daemon.jar            daemon.sh      makebase.bat   setclasspath.sh   startup.bat  tomcat-native.tar.gz  version.bat
catalina.bat        ciphers.sh   configtest.bat                digest.bat     makebase.sh    shutdown.bat      startup.sh   tool-wrapper.bat      version.sh
[root@ip-192-15-28-52 bin]# ./catalina.sh start
Using CATALINA_BASE:   /opt/apache-tomcat-9.0.111
Using CATALINA_HOME:   /opt/apache-tomcat-9.0.111
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.111/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/apache-tomcat-9.0.111/bin/bootstrap.jar:/opt/apache-tomcat-9.0.111/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[root@ip-192-15-28-52 bin]# cd ..
[root@ip-192-15-28-52 apache-tomcat-9.0.111]# cd webapps/
[root@ip-192-15-28-52 webapps]# ls
ROOT  docs  examples  host-manager  manager
[root@ip-192-15-28-52 webapps]# curl -O https://s3-us-west-2.amazonaws.com/studentapi-cit/student.war
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 89423  100 89423    0     0  75897      0  0:00:01  0:00:01 --:--:-- 75910
[root@ip-192-15-28-52 webapps]# ls
ROOT  docs  examples  host-manager  manager  student.war
[root@ip-192-15-28-52 webapps]# cd ..
[root@ip-192-15-28-52 apache-tomcat-9.0.111]# cd lib
[root@ip-192-15-28-52 lib]# ls
annotations-api.jar  catalina-storeconfig.jar  el-api.jar      jsp-api.jar          tomcat-coyote.jar   tomcat-i18n-es.jar  tomcat-i18n-pt-BR.jar  tomcat-jni.jar        websocket-api.jar
catalina-ant.jar     catalina-tribes.jar       jasper-el.jar   servlet-api.jar      tomcat-dbcp.jar     tomcat-i18n-fr.jar  tomcat-i18n-ru.jar     tomcat-util-scan.jar
catalina-ha.jar      catalina.jar              jasper.jar      tomcat-api.jar       tomcat-i18n-cs.jar  tomcat-i18n-ja.jar  tomcat-i18n-zh-CN.jar  tomcat-util.jar
catalina-ssi.jar     ecj-4.20.jar              jaspic-api.jar  tomcat-coyote-ffm.jar tomcat-i18n-de.jar tomcat-i18n-ko.jar  tomcat-jdbc.jar        tomcat-websocket.jar
[root@ip-192-15-28-52 lib]# curl -O https://s3-us-west-2.amazonaws.com/studentapi-cit/mysql-connector.jar
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100  983k  100  983k    0     0   528k      0  0:00:01  0:00:01 --:--:--  528k
[root@ip-192-15-28-52 lib]# ls
annotations-api.jar  catalina-storeconfig.jar  el-api.jar      jsp-api.jar          tomcat-coyote-ffm.jar  tomcat-i18n-de.jar  tomcat-i18n-ko.jar     tomcat-jdbc.jar       tomcat-websocket.jar
catalina-ant.jar     catalina-tribes.jar       jasper-el.jar   mysql-connector.jar  tomcat-coyote.jar      tomcat-i18n-es.jar  tomcat-i18n-pt-BR.jar  tomcat-jni.jar        websocket-api.jar
catalina-ha.jar      catalina.jar              jasper.jar      servlet-api.jar      tomcat-dbcp.jar        tomcat-i18n-fr.jar  tomcat-i18n-ru.jar     tomcat-util-scan.jar
catalina-ssi.jar     ecj-4.20.jar              jaspic-api.jar  tomcat-api.jar       tomcat-i18n-cs.jar     tomcat-i18n-ja.jar  tomcat-i18n-zh-CN.jar  tomcat-util.jar
[root@ip-192-15-28-52 lib]#
```

then inside conf folder opend file contex.xml and add endpoint username and password of the DB instance.

vim contex.xml

username="admin" password="roshni12345" endpoint.

```
root@ip-192-15-28-52:/opt/apache-tomcat-9.0.111/conf
<?xml version="1.0" encoding="UTF-8"?>
<!--
  Licensed to the Apache Software Foundation (ASF) under one or more
  contributor license agreements.  See the NOTICE file distributed with
  this work for additional information regarding copyright ownership.
  The ASF licenses this file to You under the Apache License, Version 2.0
  (the "License"); you may not use this file except in compliance with
  the License.  You may obtain a copy of the License at

      http://www.apache.org/licenses/LICENSE-2.0

  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License.
-->
<!-- The contents of this file will be loaded for each web application -->
<Context>

    <!-- Default set of monitored resources. If one of these changes, the    -->
    <!-- web application will be reloaded.                                   -->
    <WatchedResource>WEB-INF/web.xml</WatchedResource>
    <WatchedResource>WEB-INF/tomcat-web.xml</WatchedResource>
    <WatchedResource>${catalina.base}/conf/web.xml</WatchedResource>

    <!-- Uncomment this to disable session persistence across Tomcat restarts -->
    <!--
    <Manager pathname="" />
    -->
    <Resource name="jdbc/TestDB" auth="Container" type="javax.sql.DataSource"
    maxTotal="500" maxIdle="30" maxWaitMillis="1000"
    username="admin" password="123456/8" driverClassName="com.mysql.jdbc.Driver"
    url="jdbc:mysql://endpoint:3306/studentapp?useUnicode=yes&amp;characterEncoding=utf8"/>

</Context>
~
~
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~
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~
~
~
-- INSERT --                                                          35,1          All
```
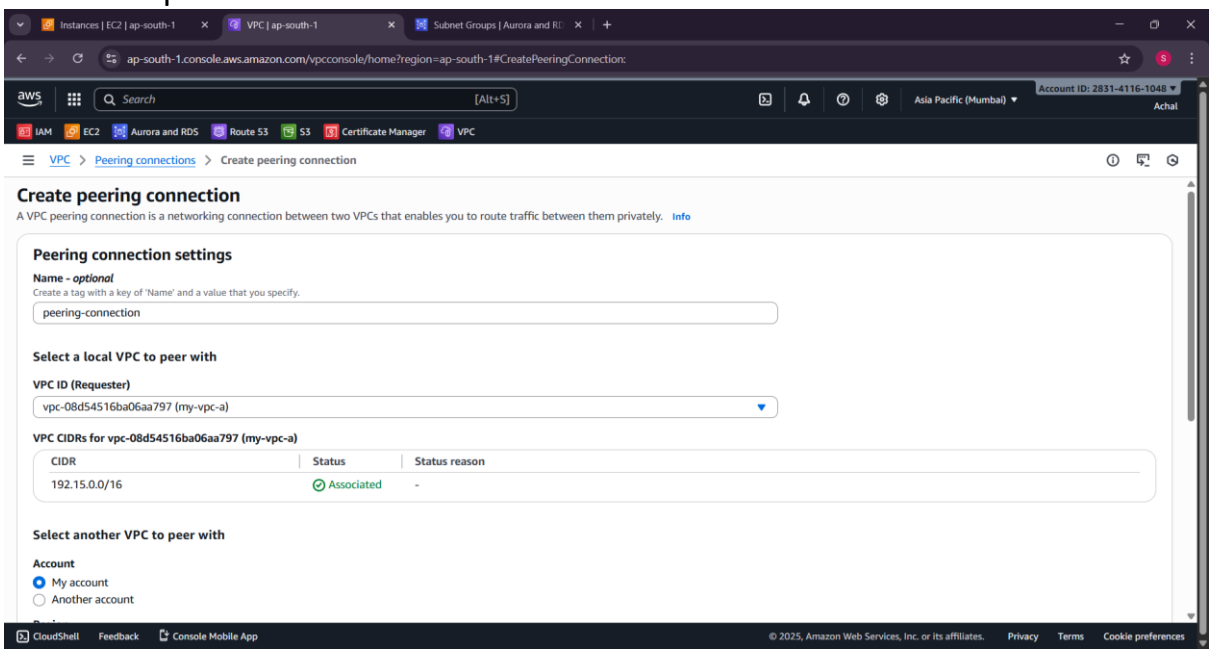
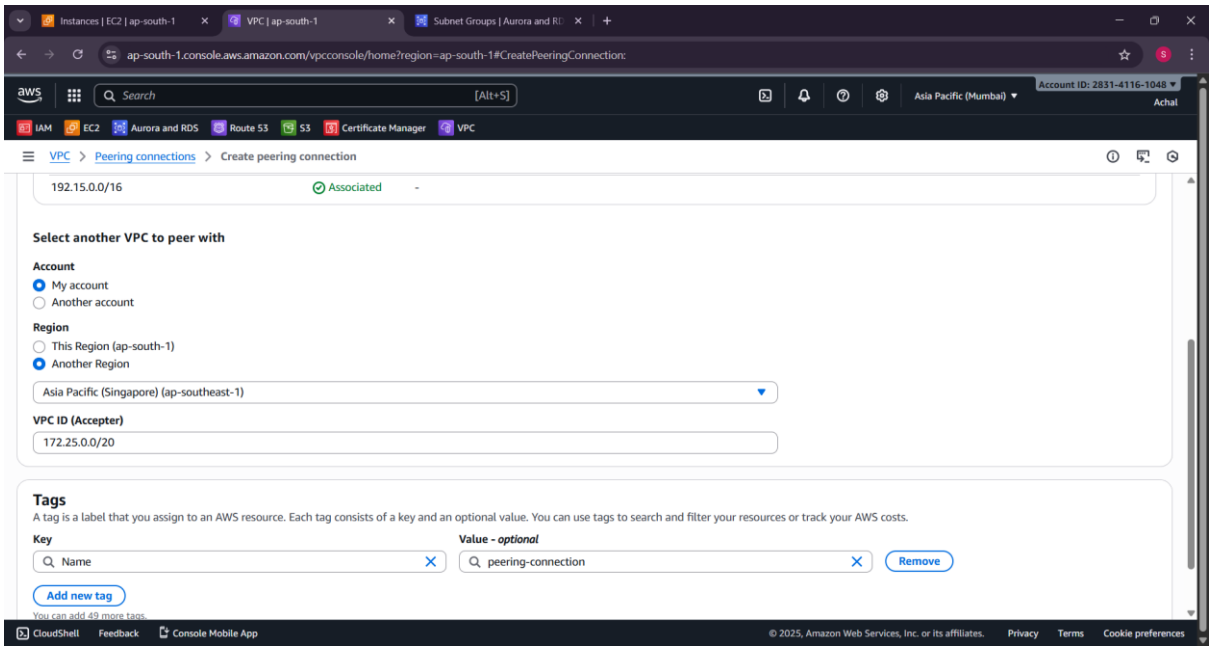
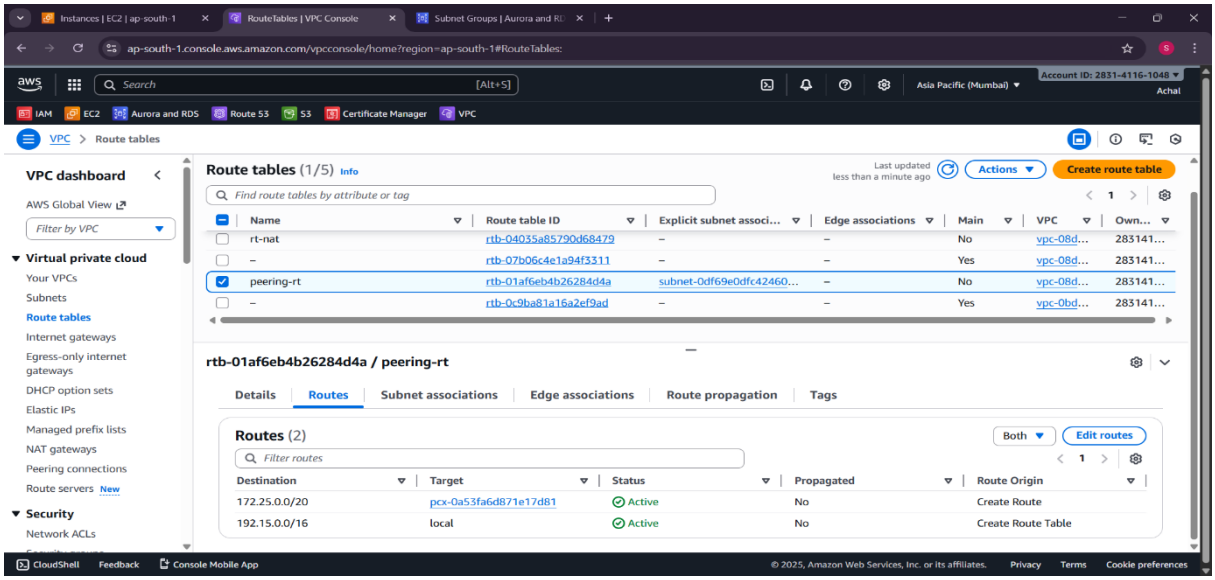
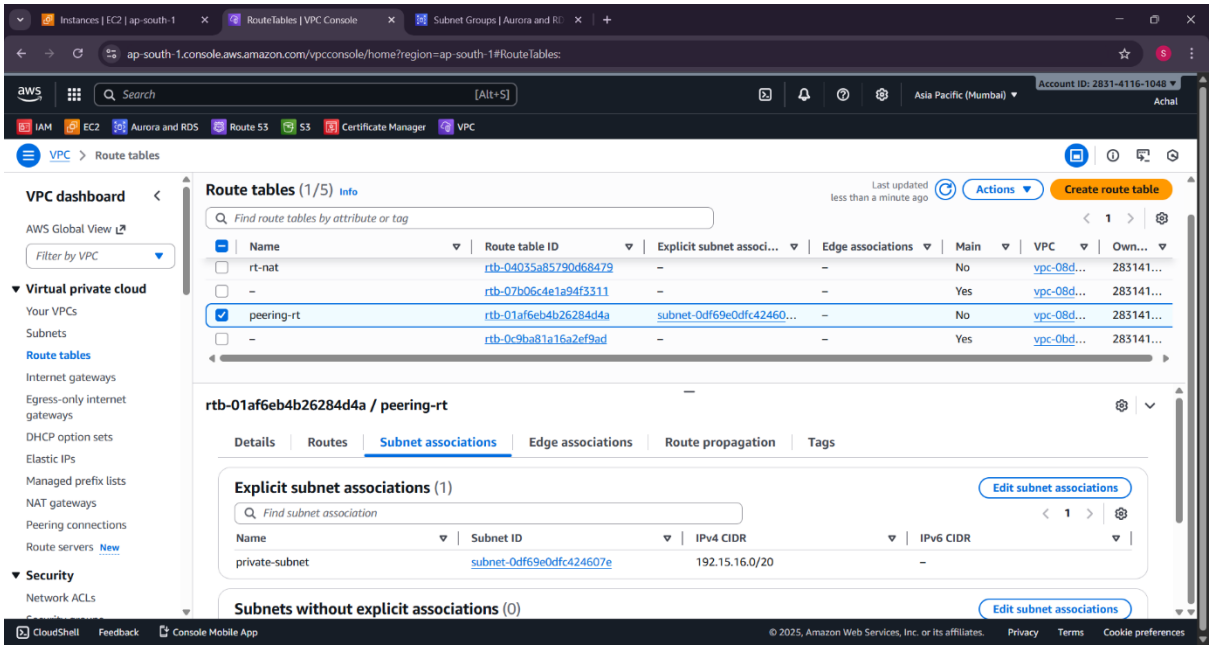

-step 17: Create a peering connection in the VPCa is a requester VPC and VPCb is the acceptor.



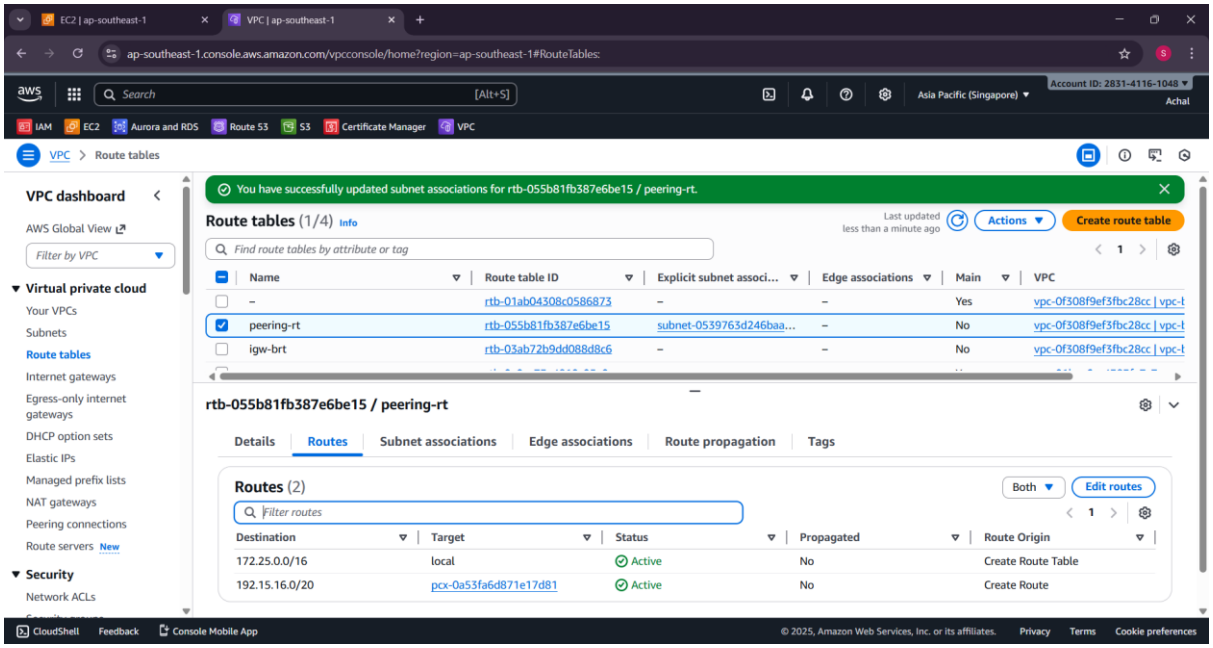


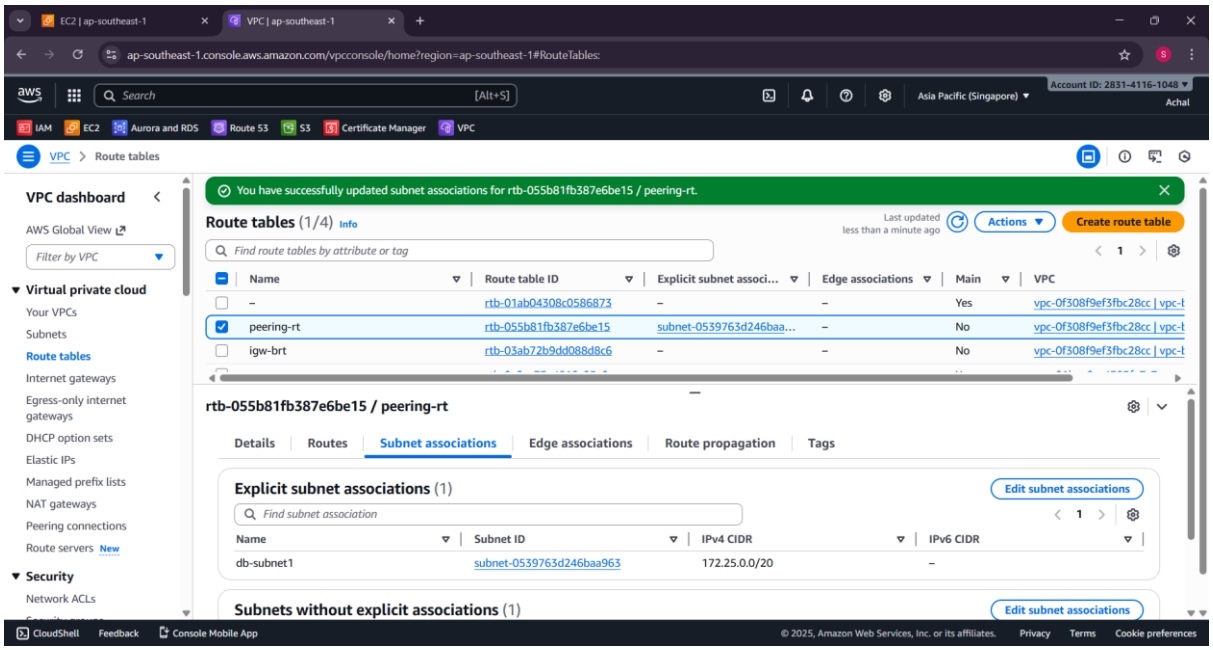
-step 18: create route table in the VPCa for VPC peering in the destination mention subnet CIDR of the VPCb subnet and choose peering connection and associate private subnet.





-step 19: in the VPCb accept that request and create route table and in the destination mention subnet CIDR of VPCa private subnet and associate subnet of VPCb. Allow All ICMP IPV4 In the security group of VPCa And VPCb. It helps to Communication and ping.





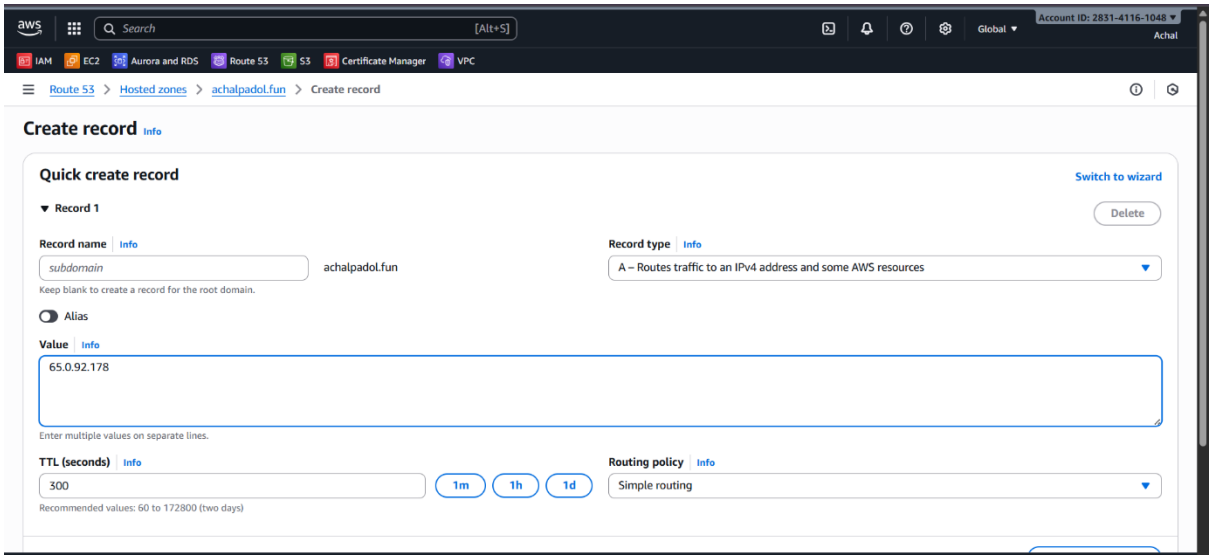
-Step 20: see result all register student are stored in the database.

Register Student

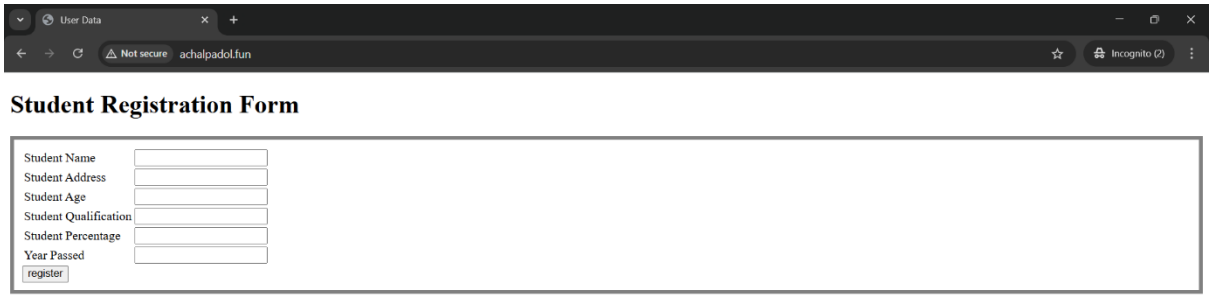
Students List

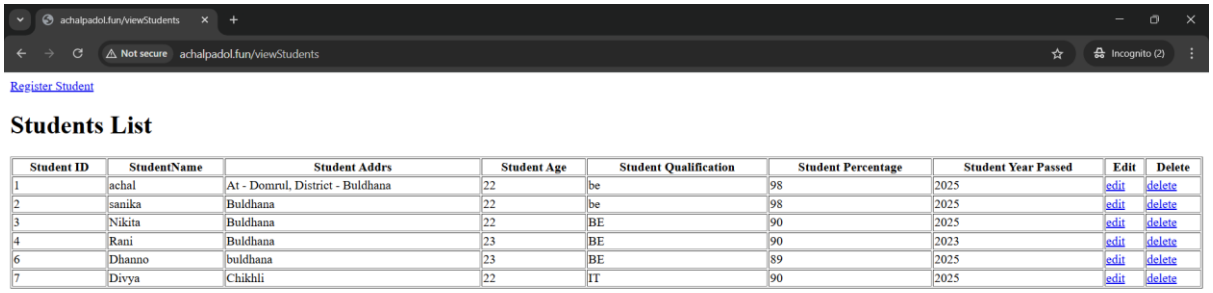
Student ID	StudentName	Student Addrs	Student Age	Student Qualification	Student Percentage	Student Year Passed	Edit	Delete
1	achal	At - Domrul, District - Buldhana	22	be	98	2025	edit	delete
2	sanika	Buldhana	22	be	98	2025	edit	delete
3	Nikita	Buldhana	22	BE	90	2025	edit	delete

-Step 21: In Route 53 service create record.



-Step 22: Now hit by using domain name.





Student ID	StudentName	Student Addrs	Student Age	Student Qualification	Student Percentage	Student Year Passed	Edit	Delete
1	achal	At - Domrul, District - Buldhana	22	be	98	2025	edit	delete
2	sanika	Buldhana	22	be	98	2025	edit	delete
3	Nikita	Buldhana	22	BE	90	2025	edit	delete
4	Rani	Buldhana	23	BE	90	2023	edit	delete
6	Dhanno	buldhana	23	BE	89	2025	edit	delete
7	Divya	Chikhli	22	IT	90	2025	edit	delete